

Endocrine Disruptors and Pollutants

Genetic sensitivity to air pollution, benzene, aromatic hydrocarbons, phthalates and bisphenol A, pesticides and glyphosate, and metals and metalloids

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. In benzene, NQO1 is intact (P187S G/G, without the function loss allele), such as CYP2E1 and EPHX1: it is the lowest toxicity genetic profile. In air pollution, NRF2 (-617) and CYP1A1 do not have risk alleles, and the predisposition to asthma is low (percentile 3.6); COPD (38.1) and lung cancer (62.6) are typical. In pesticides, butyrylcholinesterase (BCHE) is normal, Parkinson's disease is typical (35.1) and non-Hodgkin's lymphoma is low (percentile 1.1).

To be considered. PON1 has the lowest protection profile against organophosphates: Q192/Q192 and **-108 T/T** (less paraoxonase in blood). NAT2 is slow (A/A): it eliminates the aromatic amines of smoke and exhaust gases. There is 1 risk allele in **GSTP1** (I105V), **AS3MT** (arsenic) and **UGT1A1*28** (glucuronidation of BPA). MPO has the common genotype of high activity (without the benzene protective allele).

What does it mean? The genetic sensitivity to air pollutants and benzene is low to medium; the greatest vulnerability read is in organophosphate pesticides (PON1) and aromatic amines (NAT2). For phthalates, BPA and glyphosate count little: what decides is exposure.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 20 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 16 of the chip, 4 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 9 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 5 PGS Catalog scores calculated locally with 931.939 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 5 loci, 5 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

High percentile : higher risk. These diseases are associated with air pollution (COPD, asthma, lung cancer) or exposure to pesticides (Parkinson's disease, non-Hodgkin's lymphoma), but scores measure the total genetic risk and not the interaction with exposure. There are no polygenic scores for the sensitivity to pollutants.

Diseases related to air pollution

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Chronic obstructive pulmonary disease (COPD)	PGS001332 (Tanigawa 2022, 482 variants)	392 482 (weighted coverage 82%)	38,1	TIPIC	typical
Asthma	PGS001849 (Privé 2022, 22.7 thousand variants)	21.984 22.679 (weighted coverage 97%)	3,6	MENOS PROVABLE	favourable
Lung cancer (genomic)	PGS004691 (Hu 2024, LDpred2, 950 thousand variants)	909.458 950.353 (weighted coverage 96%)	62,6	TIPIC	typical

Pesticides-related diseases

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Parkinson's disease	PGS000902 (Nalls 2019, 90 variants)	87 90 (weighted coverage 96%)	35,1	TIPIC	typical
Non-Hodgkin lymphoma	PGS000080 (Graff 2021, 19 variants)	18 19 (weighted coverage 96%)	1,1	MENOS PROVABLE	favourable

4. Key variants by topic

Variants of detoxification enzymes and antioxidant defense with described effect on exposure to pollutants. When the effect depends on the compound or is inconsistent, judgment is "without judgment".

Air pollution and aromatic hydrocarbons (PAH)

Thin particles (PM_{2.5}), ozone and polycyclic aromatic hydrocarbons (PAH) of traffic, smoke and grilles enter the lung and generate oxidative stress and inflammation. CYP1A1, connected by the AHR receptor, activates PAH in reactive forms; GSTP1 and NRF2 neutralize them. The deletions of GSTM1 and GSTT1 genes, also important, do not read on a chip.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NFE2L2 (NRF2)	rs6721961	-617	G/G	imputation DR2=1.00	A (0 copy/s)	favourable	-617C>A (gene A): less NRF2, the switch of lung antioxidant defenses; associated with more lung injury with pollution and tobacco	Moderate
GSTP1	rs1695	I105V	A/G	chip	G (1 copy/s)	unfavourable (light)	I105V: lower conjugation of PAH and epoxides; associated with more effect of air pollution on asthma and lung function	Moderate
CYP1A1	rs1048943	I462V	T/T	chip	G (0 copy/s)	favourable	I462V (G of the gene = Val462): CYP1A1 more active in the activation of PAH (smoker, exhaust gas, grilled)	Weak
CYP1A1	rs4646903	MspI	A/A	imputation DR2=0.99	C (0 copy/s)	favourable	MspI (m2): CYP1A1 more inducible	Weak
AHR	rs2066853	R554K	G/G	chip	A (0 copy/s)	without judgment	R554K: hydrocarbon receptor (dioxins, PAH); small functional effect	Weak
SOD2	rs4880	V16A	G/G	chip	G (2 copy/s)	without judgment	V16A (G = Ala16): more input of superoxide dismutase in mitochondria; effect on oxidative stress of pollution is inconsistent	Weak

Benzene and solvents

Benzene (gasolin, exhaust gases, tobacco smoke, solvents) is activated in the liver by CYP2E1 and transformed into quinones toxic by myeloperoxidase (MPO). NQO1 undoes these quinones. The total lack of NQO1 is the strongest genetic factor of benzene toxicity to blood.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NQO1	rs1800566	P187S	G/G	chip	A (0 copy/s)	favourable	P187S (NQO1*2): without quinone reductase activity in homozygosity; the strongest genetic factor of hematological toxicity of benzene	Strong
MPO	rs2333227	-463	C/C	imputation DR2=1.00	A (0 copy/s)	neutral	-463G>A: less myeloperoxidase, which transforms benzene metabolites into toxic quinones	Moderate
CYP2E1	rs2031920	c1/c2	C/C	chip	T (0 copy/s)	favourable	c2 (RsaI): CYP2E1 that activates benzene and other solvents; inconsistent effect	Weak
EPHX1	rs1051740	Y113H	T/T	chip	C (0 copy/s)	favourable	Y113H (C = His113): slower epoxide hydrolase (benzene and PAH)	Weak

Bisphenol A, phthalates and hormonal disruptors

Bisphenol A (BPA) and plastic phthalates, cans and cosmetics mimic or block hormones. The body eliminates them in a few hours, especially by glucuronidation (UGT2B15 and UGT1A1), so continuous exposure weighs more than genetics. ESR1 is the estrogen receptor where BPA acts.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
UGT1A1 (*28, tag)	rs887829		C/T	chip	T (1 copy/s)	without judgment	UGT1A1*28 marker: less glucuronidation; BPA and some phthalate metabolites are eliminated by glucuronidation	Weak
UGT2B15	rs1902023	D85Y	A/C	chip	T (1 copy/s)	without judgment	D85Y (UGT2B15*2): the main UGT that eliminates bisphenol A; effect on the activity discussed	Weak
ESR1	rs2234693	Pvull	T/T	chip	C (0 copy/s)	without judgment	Pvull: variant of the estrogen receptor alpha, target of estrogen disruptors (BPA); small functional effect	Weak

Organophosphate pesticides and glyphosate

Organophosphate pesticides (chlorpyrifos, diazinon) are deactivated by PON1 paraoxonase and captured by butyrylcholinesterase (BCHE). Glyphosate is a different herbicide: it does not inhibit cholinesterases, it is little metabolized and leaves almost unchanged in urine. There are no robust human genetic markers of glyphosate sensitivity.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PON1	rs662	Q192R	T/T	chip	C (0 copy/s)	without judgment	Q192R (C in strand + = Arg192): the paraoxonase R192 hydrolyzes faster the chlorpyrifos-oxon and the paraoxon; Q192 is better for diazinon and sarin	Moderate
PON1	rs705379	-108	A/A	imputation DR2=0.97	T (2 copy/s)	unfavourable	-108C>T (gene T): less PON1 in blood, less defense against organophosphates	Moderate
BCHE	rs1803274	K	C/C	chip	A (0 copy/s)	favourable	Variant K (A539T): butyrylcholinesterase ~30% less active (organophosphates, carbamates and succinylcholine)	Moderate
BCHE	rs1799807	D70G	T/T	chip	G (0 copy/s)	favourable	Atypical variant (D70G): butyrylcholinesterase much less active	Strong
ABCB1	rs1045642	C3435T	A/A	chip	T (2 copy/s)	without judgment	C3435T (gene T): xenobiotic efflux transporter; associated with the interaction between pesticides and Parkinson's disease in some studies	Weak

Arsenic and aromatic amines

The arsenic of water and rice is methylated by AS3MT; incomplete methylation leaves more MMA, the most toxic metabolite. The aromatic amines of smoke, exhaust gases and some dyes are eliminated by acetylation (NAT2); slow acetylators accumulate them more.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
AS3MT	rs11191439	M287T	T/C	chip	C (1 copy/s)	unfavourable (light)	M287T (C = Thr287): less efficient arsenic methylation, with more MMA (the most toxic metabolite)	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NAT2 (tag)	rs1495741		A/A	chip	A (2 copy/s)	unfavourable	Slow acetylator (A/A): eliminates the aromatic amines of tobacco, exhaust gases and dyes	Strong

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) of the levels of metals and PFAS in the blood, oriented towards the highest level, compared to the reference population. The GWAS of arsenic, benzene and butadiene metabolites did not publish the effect allele and could not be used.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Blood metals (lead, cadmium, mercury)	4 (4)	3 de 8	4	47,6	without judgment
PFAS in blood	1 (1)	0 de 2	1	41,3	without judgment

6. Favourable

- **Benzene:** NQO1 P187S G/G (active NQO1), CYP2E1 c1/c1 (C/C) and EPHX1 Y113H T/T (normal activity). Strong for NQO1; weak in others.
- **Air pollution and PAH:** NFE2L2 -617 G/G (normal NRF2), CYP1A1 I462V T/T and MspI A/A without the highest activation alleles. **Moderate / weak.**
- **Low Asthma** (percentile 3.6); Typical COPD (38.1); typical lung cancer (62.6). **Moderate.**
- **Pesticides:** BCHE K C/C and D70G T/T absent (normal butyrylcholinesterase); low non-Hodgkin lymphoma (percentile 1.1); typical Parkinson's disease (35.1). **Moderate.**
- **Metals and PFAS in blood:** loci load in the 47.6 and 41.3 percentiles (typical). Weak (4 and 1 loci).

7. Unfavourable

- **PON1 with little protection against organophosphates:** -108 A/A in the strand + (T/T in the gene, less paraoxonase in the blood) and Q192R T/T (Q/Q, the slowest form for chlorpyrifos-oxon). **Moderate.**
- **NAT2 slow acetylator** (A/A): more exposure to aromatic amines of tobacco, exhaust gases and hair dyes; is the known genetic factor of bladder cancer in smokers. Strong for the slow phenotype.
- **GSTP1 I105V** A/G: 1 allele of lower conjugation; associated with more effect of air pollution on lung function. **Weak / moderate.**
- **AS3MT M287T** T/C: 1 less efficient arsenic methylation allele. **Weak.**
- **MPO -463** C/C: common genotype of high activity, without the allele that protects from benzene. **Weak.**
- **No judgment:** marker of UGT1A1*28 C/T (1 allele, glucuronidation a little lower than BPA and bilirubin); UGT2B15 D85Y A/C; ESR1 PvuII T/T; AHR R554K G/G; SOD2 V16A G/G (Ala/Ala); ABCB1 C3435T A/A (T/T in gene).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Oxidative stress	NRF2 -617 normal and NQO1 active; GSTP1 with lower conjugation 1	Favourable
Redox imbalance	SOD2 Ala/Ala (more mitochondrial superoxide dismutase, more hydrogen peroxide); without judgment	Neutral
DNA Damage and DNA Repair Deficit	Activated PAH (CYP1A1) and benzene (NQO1, CYP2E1) without risk alleles Slow NAT2 for aromatic amines	Neutral
Clonal hematopoiesis (CHIP)	The toxicity of benzene to the marrow is the main mechanism; Active NQO1 protects, high activity MPO	Neutral
Endocrine dysregulation	BPA and phthalates: slightly reduced glucuronidation (UGT1A1 and UGT2B15, 1 allele each); exposure weighs more than genetics	Neutral
Neuroinflammation	PON1 of low protection against organophosphates; typical Parkinson's disease (percentile 35.1)	Unfavourable (mild)
Chronic Inflammation	Low Asthma (percentile 3.6); COPD typical	Favourable
Dysbiosis	Glyphosate acts in the chiquimate pathway, present in gut bacteria; without human genetic marker	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Pesticides (low protection PON1):** washing well the fruit and vegetables and prefer biological in foods with more residues (browns, spinaches, apples, grapes, peppers). Do not apply organophosphate insecticides at home or in the garden; if necessary, wear gloves and mask. *Suggestion (not applied).*
2. **Brassica and NRF2:** Broccoli, broccoli shoots, kale and cress activate NRF2 and 2 phase enzymes. In an trial, a broccoli shoot drink increased urinary elimination of benzene by about 60%. A daily portion is added to the fiber. *Suggestion (not applied).*
3. **Benzene and aromatic amines (slow NAT2):** do not stay next to the pump during fuel supply, do not have engines or solvents in indoor spaces, avoid tobacco smoke (also passive) and prefer hair dyes without aromatic amines. *Suggestion (not applied).*
4. **Air inside the house:** HEPA filter in the room and ventilation when cooking (exhaustor out). On days of high pollution, training inside or away from moving roads. *Suggestion (not applied).*
5. **BPA and phthalates:** not heating food in plastic (microwaves, washing machine), prefer glass or stainless steel, reduce cans and heat sheet, and prefer cosmetics without perfume. Filtered water (activated coal or reverse osmosis) also reduces PFAS and some pesticides. *Suggestion (not applied).*
6. **Omega-3** In small trials, fish oil attenuated the effects of fine particles on heart variability and inflammation; maintain 3.2–4 g/day of EPA + DHA. *Suggestion (not applied).*
7. **Arsenic:** vary the grains and cook the rice in a lot of water (5–6 parts), flowing it, which removes 40–60% from the inorganic arsenic. *Suggestion (not applied).*

Not recommended: "detox" with chelators without proven intoxication; NAC supplements at high doses as protection against pollution (without proven benefit); organophosphate insecticides at home.

10. Limitations

- The deletions of GSTM1 and GSTT1 genes, important in air pollution and benzene, do not read on a SNP chip. NAT2 was read by a marker (tag), not by the complete genotype.
- **Glyphosate:** There are no robust human genetic markers; the statements about "genes of glyphosate sensitivity" are not solid. The action in the microbiome does not depend on Sérgio F genes.
- **Phthalates and BPA:** elimination is fast and daily exposure weighs much more than small differences in glucuronidation.
- The interaction studies between genes and pollutants have small samples and many unreplicated results. The scores of diseases related to pollution measure the total genetic risk, not the sensitivity to exposure.
- The GWAS of arsenic, benzene and butadiene metabolites do not publish the effect allele and did not enter the loci load; the charge of metals and PFAS has few loci.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromics, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS001332 (Tanigawa 2022), PGS001849 (Privé 2022), PGS004691 (Hu 2024), PGS000902 (Nalls 2019) and PGS000080 (Graff 2021). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Rothman 1997 and Lan 2004 (NQO1, MPO and benzene), Marzec 2007 (NRF2), Romieu 2006 and Minelli 2011 (GSTP1 and pollution), Costa 2013 (PON1), Agusa 2011 (AS3MT), Egner 2014 (breasts of broccoli and benzene) and the monographs of IARC on benzene, glyphosate and outdoor air pollution. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Air pollution and aromatic hydrocarbons (PAH)	NFE2L2 (NRF2)	rs6721961	G/G	imputation DR2=1.00	A	0	complement (strand)	without the risk allele
Air pollution and aromatic hydrocarbons (PAH)	GSTP1	rs1695	A/G	chip	G	1	strand +	1 copy of risk allele
Air pollution and aromatic hydrocarbons (PAH)	CYP1A1	rs1048943	T/T	chip	G	0	complement (strand)	without the risk allele
Air pollution and aromatic hydrocarbons (PAH)	CYP1A1	rs4646903	A/A	imputation DR2=0.99	C	0	complement (strand)	without the risk allele
Air pollution and aromatic hydrocarbons (PAH)	AHR	rs2066853	G/G	chip	A	0	strand +	0 copy of the effect allele
Air pollution and aromatic hydrocarbons (PAH)	SOD2	rs4880	G/G	chip	G	2	strand +	2 copy of the effect allele
Benzene and solvents	NQO1	rs1800566	G/G	chip	A	0	strand +	without the risk allele
Benzene and solvents	MPO	rs2333227	C/C	imputation DR2=1.00	A	0	complement (strand)	without the protective allele (common genotype)
Benzene and solvents	CYP2E1	rs2031920	C/C	chip	T	0	strand +	without the risk allele
Benzene and solvents	EPHX1	rs1051740	T/T	chip	C	0	strand +	without the risk allele
Bisphenol A, phthalates and hormonal disruptors	UGT1A1 (*28, tag)	rs887829	C/T	chip	T	1	strand +	1 copy of effect allele
Bisphenol A, phthalates and hormonal disruptors	UGT2B15	rs1902023	A/C	chip	T	1	complement (strand)	1 copy of effect allele
Bisphenol A, phthalates and hormonal disruptors	ESR1	rs2234693	T/T	chip	C	0	strand +	0 copy of the effect allele
Organophosphate pesticides and glyphosate	PON1	rs662	T/T	chip	C	0	strand +	0 copy of the effect allele
Organophosphate pesticides and glyphosate	PON1	rs705379	A/A	imputation DR2=0.97	T	2	complement (strand)	2 risk allele copies
Organophosphate pesticides and glyphosate	BCHE	rs1803274	C/C	chip	A	0	complement (strand)	without the risk allele
Organophosphate pesticides and glyphosate	BCHE	rs1799807	T/T	chip	G	0	complement (strand)	without the risk allele
Organophosphate pesticides and glyphosate	ABCB1	rs1045642	A/A	chip	T	2	complement (strand)	2 copy of the effect allele
Arsenic and aromatic amines	AS3MT	rs11191439	T/C	chip	C	1	strand +	1 copy of risk allele
Arsenic and aromatic amines	NAT2 (tag)	rs1495741	A/A	chip	A	2	strand +	2 risk allele copies